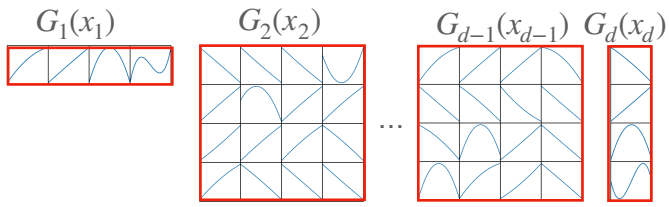
A functional tensor train by recursive Gaussian elimination

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FTT-qTensCUR

Abstract: High-dimensional function approximation is a central challenge in scientific computing, and functional tensor train (FTT) decompositions offer a practical way to break the curse of dimensionality by representing the multivariate function or *ctensor* (continuous tensor, inspired by [1]) $f : [-1,1]^d \to \mathbb{R}$ as a chain of low-dimensional cores.

$f(x_1, \ldots, x_d) \approx$ 

The (discrete) tensor train format offers a remedy to this by giving a way to represent tensors with $\mathcal{O}(n^d)$ entries with only $\mathcal{O}(n\,d\,r^2)$. A natural idea is thus: discretize a matrix on a grid and and compress it using Tensor Train. This, however, requires $\mathcal{O}(n^d)$ function evaluations, and if you want high accuracy, we will need n to be high, a prohibitive expense even for $d \geq 3$.

This raises the question: how can we form a functional tensor train without evaluating the function on a grid? We present such an algorithm along with a convergence guarantee. As far as we know, this is the first of its kind for $d \geq 3$. We show that the number of pivoting steps required per stage to achieve L^2 accuracy of ε is

$$r_* = \mathcal{O}(\log(1/\varepsilon) + d\,\log d).$$

Further, we provide a practical version of the algorithm which is able to achieve this convergence.

Algorithms

Algorithm 1 qTensCUR with complete LU pivoting

Require: $f = [f_1(x,\boldsymbol{y}) \cdots f_n(x,\boldsymbol{y})] : [-1,1]^d \to \mathbb{R}^n$, target rank r

Initialize residual $e_1(i,x,\boldsymbol{y}) = f_i(x,\boldsymbol{y})$
for $k = 1$ to r **do**
  $(i_k, x_k, \boldsymbol{y}_k) \leftarrow \arg\max_{i,x,\boldsymbol{y}} \left| e_k(i,x,\boldsymbol{y}) \right|$ (Complete pivoting)
  if $|e_k(i_k,x_k,\boldsymbol{y}_k)| <$tol **then** terminate.
  $e_{k+1}(i,x,\boldsymbol{y}) \leftarrow e_k(i,x,\boldsymbol{y}) - \frac{e_k(i,x,\boldsymbol{y}_k)}{e_k(i_k,x_k,\boldsymbol{y}_k)} e_k(i_k,x_k,\boldsymbol{y})$
end for
Form cross matrix $S \in \mathbb{R}^{r\times r}$ with entries $[S]_{j,k} = f_{i_j}(x_j, \boldsymbol{y}_k)$.
Define $U := S^{-1}$
Form $C(x) : [-1,1] \to \mathbb{R}^{n\times r}$ with entries $[C(x)]_{j,k} = f_j(x, \boldsymbol{y}_k)$.
Form $Y(\boldsymbol{y}) : [-1,1]^{d-1} \to \mathbb{R}^r$ with entries $[Y(\boldsymbol{y})]_k = f_{i_k}(x_k, \boldsymbol{y})$.
return $C(x), U, Y(\boldsymbol{y})$

Algorithm 2 QR-projected TT-qTensCUR

Require: $f : [-1,1]^d \to \mathbb{R}$, $r > 0$, Boolean `orth`

$r_0 = r_d = 1$ and $r_k = r\ \forall k \in \{1, \ldots, d-1\}$
$F_1(x_1, \ldots, x_d) := [f(x_1, \ldots, x_d)] : [-1,1]^d \to \mathbb{R}^1$
for $k = 1$ to $d-1$ **do**
  $C_k(x_k), U_k, Y(\boldsymbol{x}_{>k}) \leftarrow$ `qTensCUR`$(F_k(x_k; x_{k+1}, \ldots, x_d), r_k)$
  if `orth` is True **then**
    Orthogonalize in L^2: $C_k(x_k) = Q_k(x_k)\tilde{R}_k$
    $G_k(x_k) := Q_k(x_k) : [-1,1] \to \mathbb{R}^{r_{k-1}\times r_k}$
    $F_{k+1}(\boldsymbol{x}_{>k}) := Q_k^* F_k(\cdot, \boldsymbol{x}_{>k}) : [-1,1]^{d-k} \to \mathbb{R}^{r_k}$
  else
    $G_k(x_k) := C_k(x_k)U_k$ and $F_{k+1}(\boldsymbol{x}_{>k}) := Y_k(\boldsymbol{x}_{>k})$
  end if
end for
$G_d(x_d) = F_d(x_d)$
return cores $[G_1(x_1), \ldots, G_d(x_d)]$

Theorem. Let $f : [-1,1]^d \to \mathbb{R}$ have TT-rank $\leq r$ and $\hat{f}$ be the tensor train result of Algorithm 2 (with or without orthogonalization) ran with rank r. Then $\hat{f} = f$.

Further, for any $\varepsilon > 0$, if $f : [-1,1]^d \to \mathbb{R}$ is sufficiently analytic and bounded and we set $r = \mathcal{O}(\log(1/\varepsilon) + d\,\log d)$ then $\hat{f}$ obtained from running algorithm 2 with orthogonalization satisfies $\|f - \hat{f}\|_2 \leq \varepsilon$.

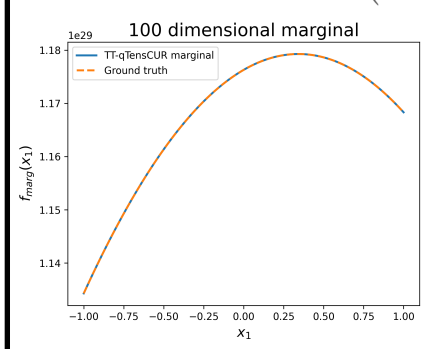
Remarks. Algorithm 2 with orthogonalization requires $\mathcal{O}(r^d)$ function evaluations and without it requires $\mathcal{O}(r\,d)$. In particular, we have provable convergence for a Chebfun3-like algorithm that extends to higher dimensions.

Proof sketch

- Under the assumption of sufficient analyticity, we know that Algorithm 1 converges geometrically in r following the argument from [1].
- Letting $E_k := (I - Q_k Q_k^*)F_k$, this leads to $\|E_k\|_2 \leq \mathcal{O}\left(\sqrt{r_{k-1}}C^{-r_k}\right)$ for for some constant C (dependent on the analyticity of the function and pivoting method used).
- The final L^2-error is bounded by the sum of these errors. That is $\|f - \hat{f}\|_2^2 \leq \sum_{k=1}^{d-1} \|E_k\|_2^2$.
- Boundedness and analyticity propagates through Algorithm 2. That is, if we have $F_k \leq M$ then $F_{k+1} \leq \sqrt{r_{k-1}}M$ and all the F_k are analytic.
- Putting this together yields $\|f - \hat{f}\|_2 \leq \mathcal{O}\left(\sqrt{d-2}\,r^{(d-1)/2}C^{-r}\right)$.
- The choice for r gives the desired bound.

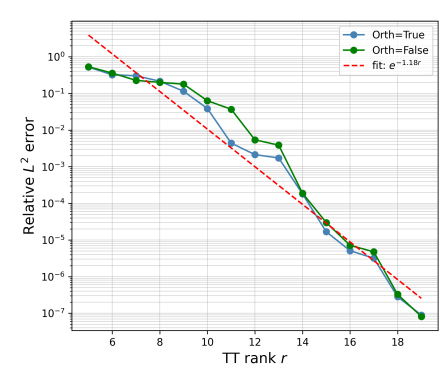
Numerical results

$$f(\boldsymbol{x}) = \cos(\boldsymbol{w}^\top \boldsymbol{x}) + \exp\left(-\frac{1}{2}\left|S^{-1}(\boldsymbol{x} - \boldsymbol{c})\right|^2\right)$$



Took 150 seconds and 19,387,075 function evaluations. The TT grid method would only see the constant basis function for a similar number of evaluations.

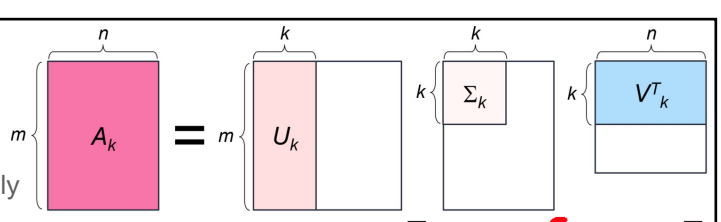
$$f(x_1, x_2, x_3, x_4) = \cos\left((x_1 + x_2 + x_3 + x_4)^2\right)$$



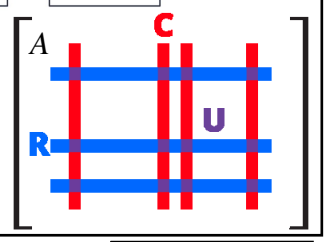
Background

Low rank matrix decompositions

Given $A \in \mathbb{R}^{m\times n}$ the *singular value decomposition* (SVD) writes $A = U\Sigma V^\top$ where $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ have orthonormal columns and $\Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_{\min\{m,n\}}) \in \mathbb{R}^{m\times n}$ with $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$, the *singular values*. Keeping only the first k singular values gives the best rank-k truncation $A \approx A_k = U_k\Sigma_k^\top V_k^\top$ where $U_k \in \mathbb{R}^{m\times k}$, $V_k \in \mathbb{R}^{n\times k}$, and $\Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_k)$. This costs $\mathcal{O}((n+m)k)$ to store rather than $\mathcal{O}(nm)$.



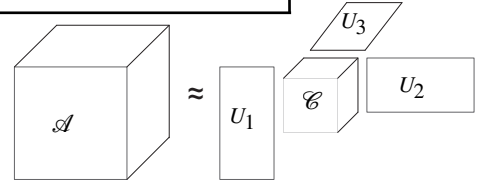
However, these factors U_r and V_r are a bit abstract. To make things interpretable, the *CUR decomposition* of A writes $A \approx CUR$ where C is k columns of A, R is k rows of A, and U^{-1} is their intersection.



Low rank tensor decompositions

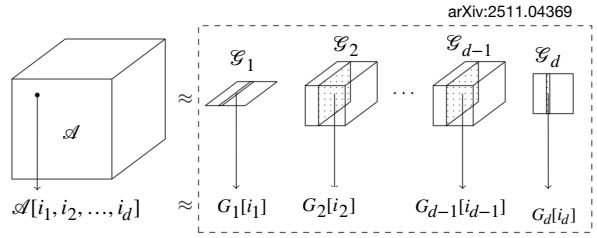
Attempting to extend the SVD to a d-way tensor $\mathcal{A} \in \mathbb{R}^{n\times\cdots\times n}$ gives rise to the *Tucker decomposition* $\mathcal{A} \approx \mathcal{C} \times_1 U_1 \times_2 \cdots \times_d U_d$, where $\mathcal{A} \in \mathbb{R}^{k\times\cdots\times k}$ and $U_j \in \mathbb{R}^{n\times k}$, bringing the storage complexity from $\mathcal{O}(n^d)$ to $\mathcal{O}(k^d)$.

arXiv:2010.10131



To avoid this exponential blow up, we consider the *tensor train (TT) decomposition*, where we write $\mathcal{A}[i_1, \ldots, i_d] \approx G_1[i_1]G_2[i_2]\cdots G_d[i_d]$ with $G_j \in \mathbb{R}^{k_{j-1}\times n\times k_j}$ where $k_0 = k_d = 1$ and $k_j = k$ for $j = 1, \ldots, d-1$. This requires only $\mathcal{O}(k^2 n\,d)$ parameters.

arXiv:2511.04369



Extending to the continuum

Discrete $m \gg k$	Skinny matrix $A \in \mathbb{R}^{m\times k}$	Skinny tensor $\mathcal{A} \in \mathbb{R}^{m\times\cdots\times m\times k}$ (d such m's)	CUR of a skinny tensor $\mathcal{A} \approx \mathcal{C}U\mathcal{R}$ where $\mathcal{C} \in \mathbb{R}^{m\times k\times r}$, $U \in \mathbb{R}^{r\times r}$, $\mathcal{R} \in \mathbb{R}^{r\times m\times\cdots\times m\times k}$	TT of a tensor $\mathcal{A} \in \mathbb{R}^{n\times\cdots\times n}$ $\mathcal{A}[i_1, \ldots, i_d] \approx G_1[i_1]\cdots G_d[i_d]$
Continuous	Quasimatrix $f : [-1,1] \to \mathbb{R}^k$	Quasitensor (qTens) $f : [-1,1]^d \to \mathbb{R}^k$	CUR of a qTens $f(x,\boldsymbol{y}) \approx C(x)UR(\boldsymbol{y})$ where $C : [-1,1] \to \mathbb{R}^{k\times r}$ and $R : [-1,1]^{d-1} \to \mathbb{R}^r$	FTT of a multivariate function $f : [-1,1]^d \to \mathbb{R}$ $f(x_1, \ldots, x_d) \approx G_1(x_1)\cdots G_d(x_d)$ where $G_j : [-1,1] \to \mathbb{R}^{r_{j-1}\times r_j}$

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[4] Cortinovis, Alice, Daniel Kressner, and Stefano Massei. "On maximum volume submatrices and cross approximation for symmetric semidefinite and diagonally dominant matrices." *Linear Algebra and its Applications* 593 (2020): 251–268.
[5] Jeong, Sungwoo, and Alex Townsend. "Convergence of Pivoted Cholesky Algorithm for Lipschitz Kernels." *arXiv preprint arXiv:*2509.13582 (2025).